

Population Genetics Drill Problems

Instructions

These problems are designed to practice the core calculations in population genetics. Apply the standard equations for Hardy-Weinberg equilibrium, selection, mutation, and migration. Show all your work.

(1) Allele and Genotype Frequency Calculations

1. In a population of 500 individuals, the genotype counts at a locus are: $AA = 245$, $Aa = 210$, $aa = 45$. Calculate:
 - a. The frequency of allele A
 - b. The frequency of allele a
2. In a population, the frequency of the recessive allele (q) is 0.3. Assuming Hardy-Weinberg equilibrium, calculate:
 - a. The frequency of the dominant allele (p)
 - b. The expected frequency of homozygous dominant individuals
 - c. The expected frequency of heterozygotes
 - d. The expected frequency of homozygous recessive individuals
3. A rare autosomal dominant disorder affects 1 in 10,000 individuals in a population. Assuming Hardy-Weinberg equilibrium, what is the frequency of the disease allele?
4. In a population of 2,000 individuals, 320 individuals show a recessive trait. Assuming Hardy-Weinberg equilibrium, calculate:
 - a. The frequency of the recessive allele
 - b. The frequency of the dominant allele
 - c. The number of heterozygous individuals expected
5. At an X-linked locus in a population of mammals, the frequency of the recessive allele in males is 0.15. Assuming Hardy-Weinberg equilibrium, what is the expected frequency of:
 - a. Affected males
 - b. Affected females
 - c. Carrier females

(2) Hardy-Weinberg Equilibrium Testing

6. A locus has two alleles. Observed genotype counts: $AA = 50$, $Aa = 20$, $aa = 30$ (total $N = 100$). Calculate:

- The observed allele frequencies
- The expected genotype counts under HWE
- The chi-square value (use 1 degree of freedom, critical value = 3.84). Is the population in HWE?

7. At a locus with two alleles, the observed heterozygote frequency is 0.42, and the frequency of the dominant allele is 0.6. What is the expected heterozygote frequency under HWE? Is there an excess or deficiency of heterozygotes?

8. In a population of 1,000 individuals, the observed genotype counts are: $AA = 640$, $Aa = 320$, $aa = 40$. Calculate the expected counts under HWE and determine if the population is in equilibrium.

(3) Selection Models

9. At a locus with two alleles, the relative fitnesses are: $w_{AA} = 1.0$, $w_{Aa} = 1.0$, $w_{aa} = 0.4$. The current frequency of the recessive allele (a) is 0.2. Calculate the mean fitness of the population and the frequency of a in the next generation after selection.

10. For an overdominant locus, the fitnesses are: $w_{AA} = 0.7$, $w_{Aa} = 1.0$, $w_{aa} = 0.4$. Calculate the equilibrium frequency of the A allele.

11. For an underdominant locus, the fitnesses are: $w_{AA} = 1.0$, $w_{Aa} = 0.6$, $w_{aa} = 1.0$. Calculate the unstable equilibrium frequency of the A allele.

12. At a locus, the current frequency of the dominant allele (A) is 0.8. The fitnesses are: $w_{AA} = 1.0$, $w_{Aa} = 1.0$, $w_{aa} = 0.0$ (lethal recessive). Calculate the frequency of A in the next generation after selection.

(4) Mutation Models

13. The forward mutation rate from allele A to a is $\mu = 2 \times 10^{-5}$. The backward mutation rate from a to A is $\nu = 1 \times 10^{-6}$. Assuming no selection, calculate the equilibrium frequency of allele a .

14. A deleterious recessive allele has a selection coefficient $s = 0.3$ against homozygotes. The mutation rate from the wild-type to the deleterious allele is $\mu = 5 \times 10^{-6}$. Calculate the equilibrium frequency of the deleterious allele under mutation-selection balance.

15. A deleterious dominant allele has a selection coefficient $s = 0.2$ against heterozygotes. The mutation rate from the wild-type to the dominant allele is $\mu = 1 \times 10^{-5}$. Calculate the equilibrium frequency of the dominant allele under mutation-selection balance.

16. At a locus, the forward mutation rate from A to a is $\mu = 3 \times 10^{-5}$ and there is no backward mutation. The selection coefficient against the recessive homozygote (aa) is $s = 0.5$. Calculate the equilibrium frequency of the recessive allele.

(5) Migration Models (Continent-Island)

17. On the continent, the frequency of allele A is 0.9. On the island, the frequency of A is 0.6. The migration rate from the continent to the island is 0.08 per generation. Assuming no selection, what will be the frequency of A on the island in the next generation?

18. On the continent, the frequency of a beneficial allele is 1.0 (fixed). On the island, the frequency of the same allele is 0.3. The migration rate from the continent to the island is 0.05 per generation. Assuming no selection, what is the frequency of the allele on the island after one generation of migration?

19. On the continent, the frequency of allele A is 0.7. On the island, the current frequency of A is 0.4. The migration rate from the continent to the island is 0.10 per generation. Assuming no selection, what will be the frequency of A on the island in the next generation?

20. On the continent, the frequency of allele A is 0.95. On the island, the frequency of A is 0.45. After one generation of migration from the continent to the island (with no selection), the frequency of A on the island becomes 0.52. What was the migration rate (m)?